Full Stack Engineering

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Project Synopsis

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ENHANCING RETRIEVAL-AUGMENTED GENERATION THROUGH HYBRID RETRIEVAL STRATEGIES



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ABSTRACT

Large language models have demonstrated remarkable fluency in natural language understanding and generation, but they remain fundamentally limited by the static, parametric nature of the knowledge acquired during pre-training. They cannot access information published after their training cutoff, they have no visibility into private or organisational corpora, and when asked questions beyond their parametric knowledge they tend to produce fluent but factually incorrect output — a failure mode commonly described as hallucination. Retrieval-Augmented Generation (RAG) addresses these limitations by coupling a generative model with an external, non-parametric knowledge store: relevant passages are retrieved at query time and supplied to the model as grounding context, so that answers are conditioned on verifiable evidence rather than on memory alone. RAG has consequently become the dominant architecture for enterprise question answering, documentation assistants, legal and medical information retrieval, and conversational search.

However, the effectiveness of any RAG system is bounded almost entirely by the quality of its retrieval stage. If the correct passage is not present in the retrieved context, no amount of generative capability can recover the correct answer. Conventional RAG pipelines rely on a single retrieval paradigm, and each paradigm carries a characteristic weakness. Dense retrieval, which encodes queries and passages into a shared embedding space and ranks by vector similarity, captures paraphrase and conceptual relatedness well, but it frequently fails on exact identifiers, product codes, rare named entities, acronyms, numerical values and specialised domain terminology that were poorly represented in the embedding model's training distribution. Sparse lexical retrieval such as TF-IDF and BM25 matches surface terms precisely and handles rare tokens robustly, but it collapses under vocabulary mismatch, returning nothing useful whenever a user phrases a question differently from the source document. Both failure modes propagate directly into the generated response as omissions, partial answers or confident fabrications.

This project proposes, implements and evaluates a hybrid retrieval framework that combines complementary retrieval signals within a single, unified pipeline. Documents are first processed through a semantic chunking stage that segments text along structural and topical boundaries with a configurable overlap window, preserving contextual continuity that fixed-length chunking destroys. Each chunk is indexed twice — as a dense

embedding vector in a vector database, and as a term-frequency posting list for sparse scoring. At query time, the incoming question is optionally expanded and rewritten, dispatched in parallel to both retrievers, and the two ranked candidate lists are merged using Reciprocal Rank Fusion with a tunable weighting parameter that controls the relative influence of semantic and lexical evidence. The fused candidate set, deliberately over-retrieved to maximise recall, is then reordered by a transformer cross-encoder that jointly attends to the query and each candidate passage, producing a precision-oriented final ranking. Metadata filtering and deduplication are applied before the top-k passages are assembled into the generation prompt.

The system is implemented using an open-source sentence embedding model, a vector database with approximate nearest-neighbour indexing, a BM25 sparse index, a cross-encoder reranker and an instruction-tuned language model for generation. Evaluation is carried out on established open-domain question-answering benchmarks as well as a custom domain-specific corpus that deliberately contains identifiers, abbreviations and terminology known to challenge dense-only retrieval. Retrieval quality is measured using Recall@k, Mean Reciprocal Rank and Normalised Discounted Cumulative Gain, while end-to-end answer quality is assessed using Exact Match, token-level F1, and faithfulness and context-relevance scores computed with an automated RAG evaluation framework supplemented by human judgement on a sampled subset.

Comparative experiments against dense-only and sparse-only baselines quantify the contribution of hybrid fusion, and a systematic ablation study isolates the individual effects of semantic chunking, query expansion, fusion weighting and cross-encoder reranking. Latency and throughput are profiled at each stage so that the accuracy gains can be interpreted against their computational cost. The project aims to establish that hybrid retrieval consistently surfaces more relevant grounding context and yields more faithful, less hallucinatory answers than any single retrieval strategy, and to provide practical guidance on configuring fusion weights and reranking depth for deployment scenarios operating under real-world latency constraints.